THE FEDERAL POLYTECHNIC, MUBI

DEPARTMENT OF ELECTRICAL /ELECTRONIC ENGINEERING

ELECTRICAL POWER SYSTEM –III
EEP 326

REVIEW QUESTIONS

2019/2020
ACADEMIC SESSION

REVIEW QUESTIONS

1. In your own words, briefly explain in not more than seven lines how electricity reaches your homes from distant power generating plants.
2. Briefly explain:
    a. Advantages of interconnected power systems in terms of;
        i. Económic exchange.
        ii. Environment and new plant siting.
    b. Disadvantages of interconnected power systems in terms of;
        i. Short circuit current level.
        ii. Cost.
3. Briefly explain bus admittance matrix of power network.
4. Given the single line diagrams of power systems in fig. 1.0, obtain the bus admittance matrices by the rule of inspection;

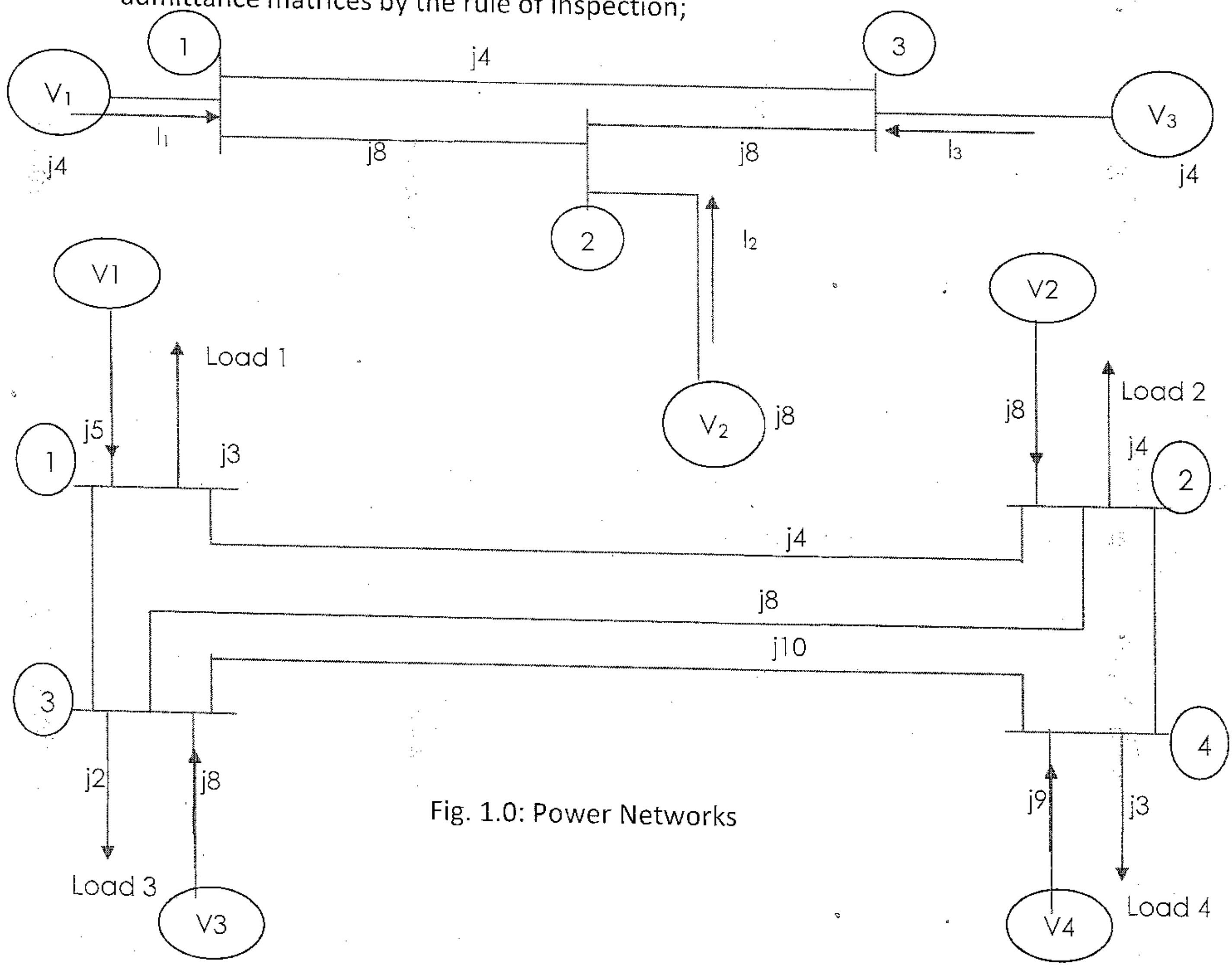

Fig. 1.0: Power Networks

5. What is power flow study?
6. What are the information that are obtained from a load flow study?
7. What is the need for load flow study?
8. What are the works involved in a load flow study?
9. What are the quantities that are associated with each bus in a power network?
10. What are the different types of buses?
11. What is voltage controlled bus?
12. What is P.Q bus?
13. What is swing bus?
14. What are the iterative methods used for solution of load flow problem?
15. Given the power network of Fig. 2.0, obtain;
    a. Y-Bus Matrix
    b. Bus voltages using Gauss-Seidel method with only four (4) iterations
    c. Power supplied by Bus (1)

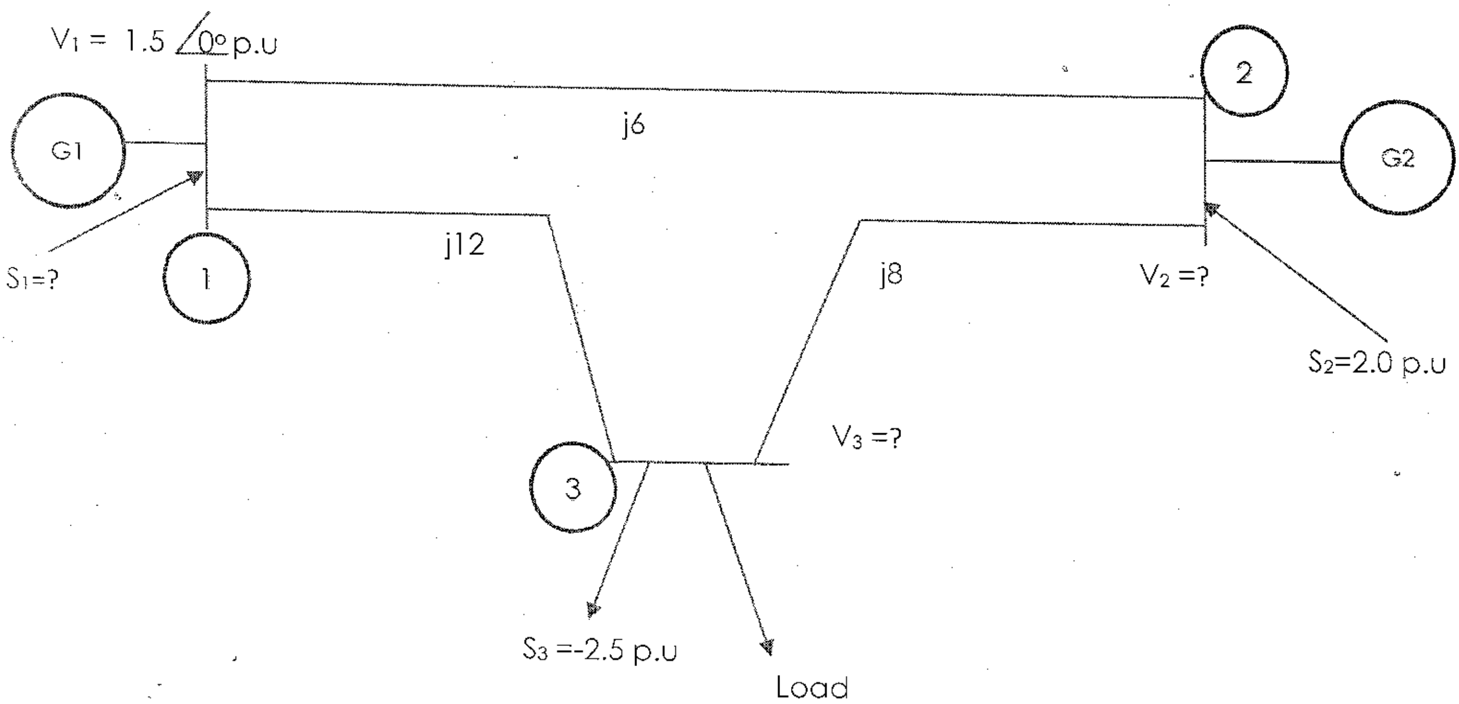

Fig. 2.0: Power Network.

16. State generator faults.
17. State transformer faults.
18. State transmission line faults.
19. What is synchronous reactance?
20. Define sub-transient reactance.

21. Define transient reactance.
22. Sketch the waveform showing sub-transient, transient and steady state periods of the fault current.
23. Sketch the waveform of the final stage of fault current of synchronous generator showing the peak values of steady state, transient, and sub-transient short circuit currents.
24. A generator rated at 50MVA, 11.8KV has a reactance of 20%. Calculate its p.u reactance for a base of 100MVA and 10KV.
25. The base KV and base MVA of a 3-phase transmission line is 33KV and 20MVA respectively. Calculate the base current and base impedance.
26. A 120MVA, 13.8KV, 3-phase 50Hz synchronous generator is operating at the rated voltage and no-load when a 3-phase fault occurs at its terminals. Its reactances per unit to the machines own base are X_{ss} =1.0 p.u, X^{I} = 0.25 p.u and X^{II} =0.12 p.u. calculate the subtransient, transient and steady state currents for the fault.
27. Find the fault current, I_F in fig. 3.0 if the pre-fault voltage at fault point is 0.97 p.u.

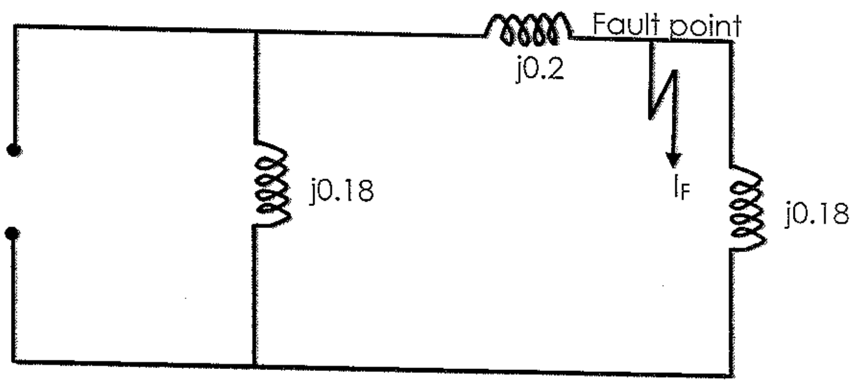

Fig.3.0: Power Network

28. What is meant by a fault?
29. State causes of fault in power network
30. List various types of shunt and series faults
31. State four types of faults on power lines and their frequency of occurrence
32. What is meant by fault calculations or fault analysis?
33. What is the need for short circuit studies or fault analysis?
34. What is meant by fault level?
35. Determine the fault level and fault current at 2.4Kv bus bar of the network of Fig. 4.0.

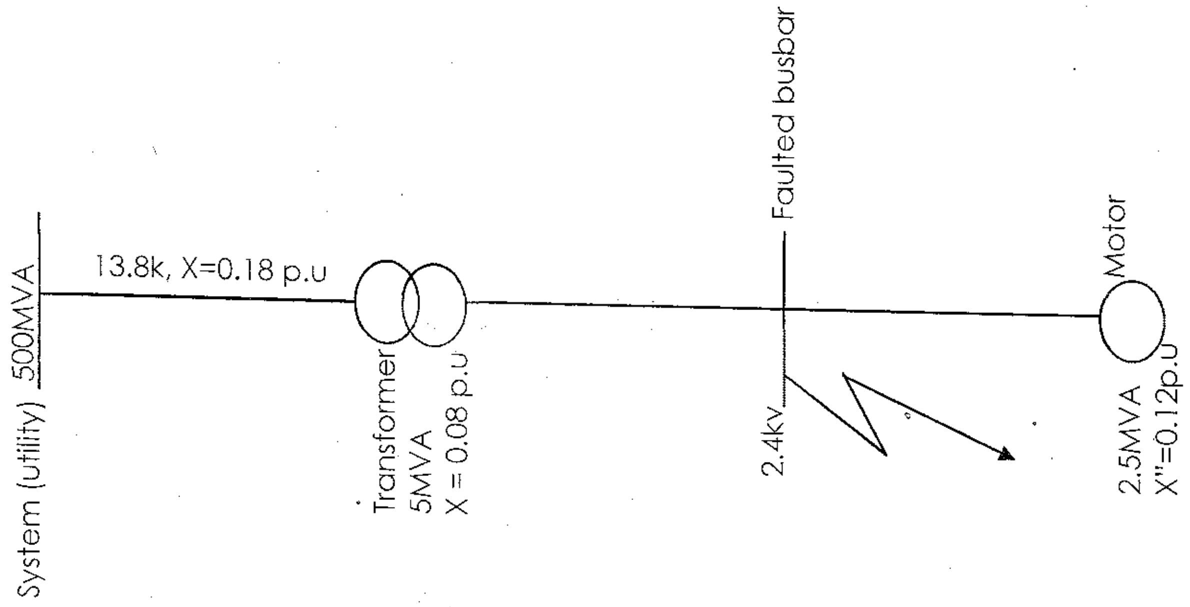

Fig. 4.0: Power Network.

36. What are symmetrical and unsymmetrical faults?
37. An 11.8kV bus bar is fed from three synchronous generators having the following ratings and synchronous reactance as follows:
    50MVA, X=0.08 p.u, 60MVA, X =0.1p.u, and 30MVA, X = 0.09 p.u. Calculate the fault MVA level and fault current if a 3-phase symmetrical fault occurs on the bus bar.
38. How are sequence components related to symmetrical components?
39. Define positive, negative and zero sequence components
40. Derive the expression for positive sequence current.
41. Derive the expression for negative sequence current
42. In a 3-phase, 4-wire system, the currents in R, Y, and B lines under unbalanced conditions of loading are as hereunder:

    $I_R = 120\angle 30^\circ$ A, $I_y = 60\angle 300^\circ$ A, $I_B = 30\angle 150^\circ$ A. Calculate the positive, negative and zero sequence currents in R-line and return current in the neutral wire.
43. Briefly explain sequence impedances of power system components.
44. Briefly explain sequence impedances of;
    i. Generators,
    ii. Transformers,
    iii. Transmission lines.
45. Use sequence impedances to drive the expression for fault current.
46. A 3-phase 10MVA, 11KV generator with solidly earthed neutral point supplies a feeder. The relevant sequence impedances of the generator and feeder are:

Sequence impedance		Generator	Feeder
Z_1		j1.3	j1.1
Z_2		j0.8	j1.1
Z_o		j0.5	j2.5

If a fault from one phase to earth occurs on the far end of the feeder of red phase, calculate the magnitude of the fault current and line to neutral voltage at the generator terminals.

47. What is sequence network of power system?
48. Draw the positive and negative sequence networks of the power elements in fig. 5.0.

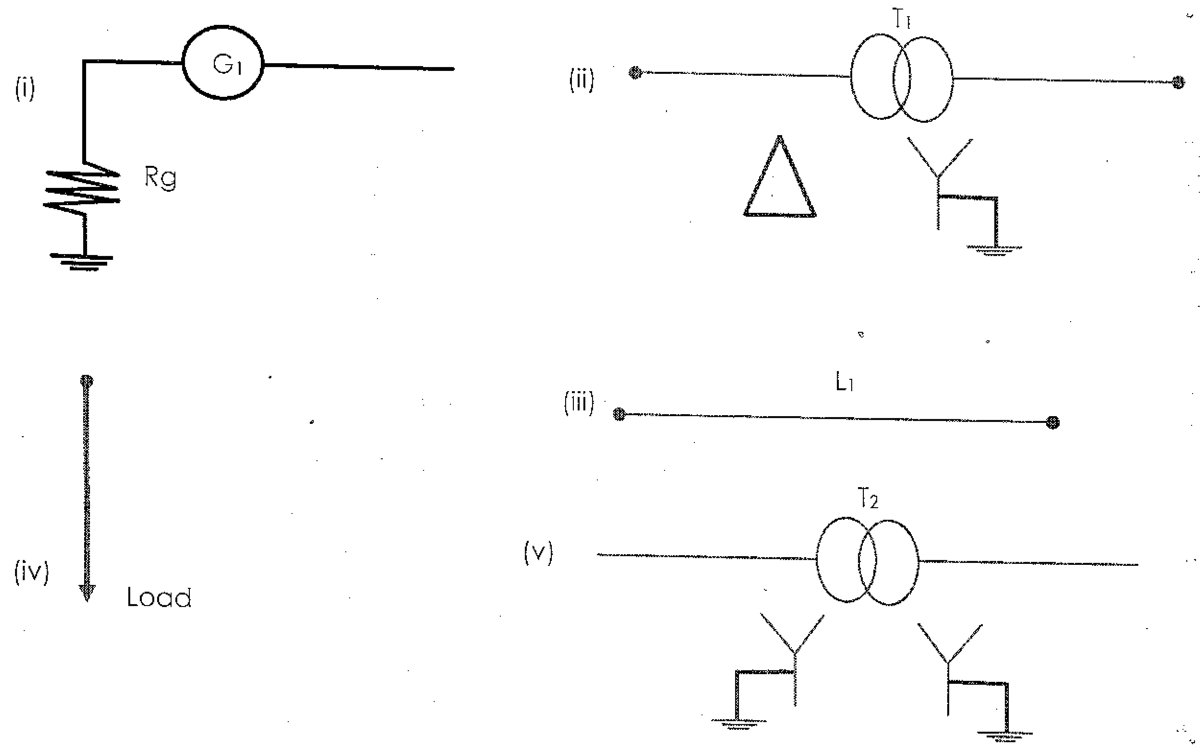

Fig. 5.0: Power Elements.

49. Construct the sequence network of the power system shown in fig 6.0

Assign 2

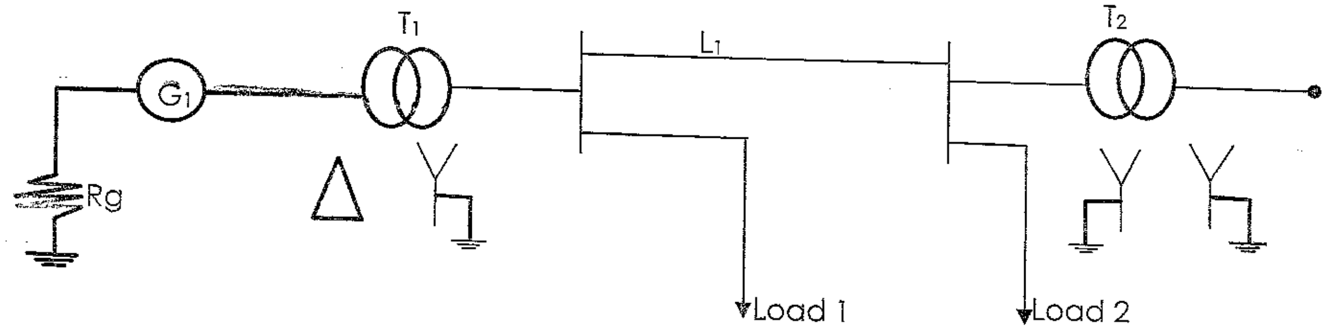

Fig. 6.0: Power System.

50. Briefly explain the following:
   a. Switch gear,
   b. Circuit breaker,
   c. Bus bar,
   d. Fuse .

Assign 2

51. Select the following:
   a. Switch gear to be installed for 33kV, 15KA, I_F =20KA
   b. Copper and Aluminum bus bars for I_F = 100KA, fault duration =5 seconds

Assign 3

52. Select fuses for norminal currents;
   a. 48A
   b. 80A
   c. 160A

53. What is corona?

54. State factors affecting corona

Briefly explain

55. State the effects of corona

56. Derive the expression for critical disruptive voltage of a line

57. A 3-phase line has conductors 2cm in diameter spaced equally 1M apart. If the dielectric strength of air is $30KV_{(MAX)}/cm$, find the critical disruptive voltage for the line. Take air density factor, δ=0.89 and irregularity factor m_o = 0.78.

58. State the causes of over-voltages on power systems.

59. What is voltage surge, sketch a labelled waveform of lightning surge voltage.

Briefly explain

60. List effects of travelling wave on a transmission system

61. Prove mathematically that the surge velocity in a uniform line is $3x10^8 m/s$

62. Deduce an expression for surge impedance

63. Briefly explain the different types of terminations of transmission lines.

64. Deduce expressions for transmitted and reflected surge voltages
65. Deduce expressions for transmitted and reflected surge currents.
66. Define reflecting factor of coefficient and transmission factor of coefficient
67. A wave travels along a 500km line terminated with a resistance of 1000Ω. The line has a resistance of 0.4Ω/Km and surge impedance of 400Ω. If the voltage at termination point after two successive reflections is 200kV, determine the amplitude of the incoming surge.
68. An overhead line with inductance and capacitance per kilometer length of 1.5mH and 0.1μF, respectively is connected in series with an underground cable having inductance and capacitance of 0.3mH/km and 0.4μF/km, respectively. Calculate the values of reflected and transmitted waves of voltage and current at the junction due to a voltage surge of 100kV traveling to the junction from the overhead line.
69. Briefly explain the methods of protection of transmission systems against lighting surges.
70. Briefly explain what happens on transmission line when it is terminated at;
    a. R =Zo,
    b. R >Zo,
    c. R<Zo,
    d. R=∞,
    e. R = 0.
71. Briefly explain the need for insulators on overhead lines
72. Draw the diagrams of the following insulators and ~~state~~ briefly explain their uses;
    a. Pin –type insulator,
    b. One unit suspension-type insulator (string insulator),
    c. Coach screw insulator.
73. State type tests of insulators.
74. Briefly explain routine tests on insulators .
75. Define string efficiency and its importance.
76. In a 33kV overhead line, there are three units in the string of insulators, if the capacitance between each insulator pin and earth is 12% of self-capacitance of each insulator, find;
    a. The distribution of voltage over the three units
    b. The string efficiency
77. Briefly explain insulation co-ordination in ~~overhead lines~~ power systems.
78. Briefly explain the principle of operation of an impulse generator.
79. Sketch and briefly explain the wave shape obtained from an impulse generator.

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